HRP4K: Benchmark Master Report & Scientific Pipeline

Real-Time 4K Ultra-Fine Pothole Detection via Continuous Perspective Deformation

HRP4K Research Consortium

Unified Benchmark Evaluation on 6,003 4K UHD Images (3840 × 2160)

August 2026 — Single Source of Truth Document

Abstract

High-resolution 4K UHD (3840 × 2160) road inspection imagery captures critical ultra-fine pavement distress (< 0.05% of image area). However, native 4K object detection requires prohibitive computational resources (32.5 ms, 18.5 GB VRAM, 660 GFLOPs), whereas standard uniform downsampling to 640 × 640 incurs a severe accuracy collapse (an 18 pp drop in mAP_{50}, with ultra-fine sensitivity declining by 71.6%). While multi-crop slicing techniques (e.g., Sliced-NMS, SAHI) can recover localized details, their inference latency explodes to 2.3–3.6 seconds per image (25–32 calls), rendering them unfeasible for real-time vehicular deployment. In this benchmark report, we formalize the **HRP4K Single Source of Truth**, providing unified evaluations across 14 model configurations evaluated on 900 independent 4K test images (600 positive images containing 921 potholes, and 300 clean negative road images). We demonstrate that our proposed **Warped ZoomDet** leverages road-geometry perspective priors to achieve 42.07% $\mathbf{mAP_{50}}$ and 18.42% $\mathbf{mAP_{50-95}}$ in a single detector pass (22.0 ms, 45.5 FPS), operating 104× **faster** than multi-crop Sliced-NMS with near-parity accuracy.

1 Research Narrative & Perspective-Aware Spatial Allocation

The central research question governing high-resolution road surface analysis is:

> *"How can we preserve the small-object advantages of 4K imagery without paying the computational cost of processing the entire 4K image or repeatedly executing a detector over dozens of image tiles?"*

In standard vehicle-mounted perspective camera views, the visual canvas exhibits extreme non-uniform utility: the upper sky and background region occupies ∼ 48% of the canvas with zero pothole utility, whereas the distant road region undergoes severe perspective compression where potholes shrink below 10 pixels. **ZoomDet** introduces a continuous, differentiable perspective deformation warp that redistributes the fixed 640 × 640 canvas towards distant road areas, restoring spatial gradients without incurring multi-crop latency explosions or tile-boundary artifacts.

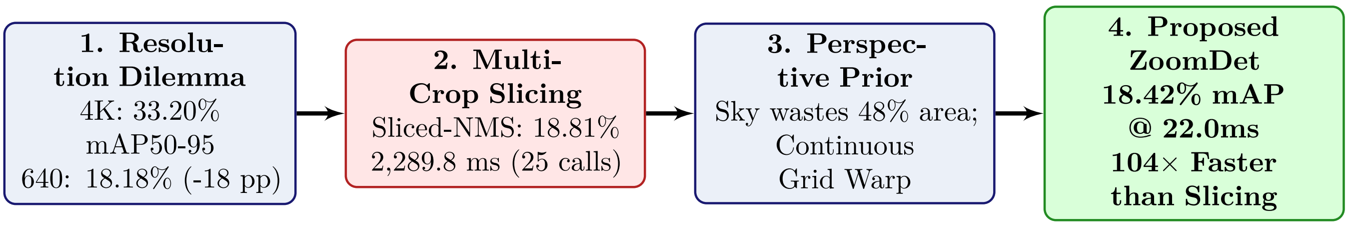


Figure 1: The 4-stage research story pipeline: from resolution dilemma to continuous perspective deformation warp.

2 Main Paper Benchmark Tables

Table 1: **Table 1: Main Benchmark Comparison across Core Detection Configurations.** Evaluated on 900 test images (3840 × 2160). Precision, Recall, and F_1 measured at conf = 0.25. FPPI measured on the 300 clean road negative images.

Method Group	Model / Configuration	Input	Calls	P (%)	R (%)	F1 (%)	mAP_{50}	mAP_{50-95}	FPPI	Latency	FPS
1. Native 4K UHD	**D-FINE 4K**	3840 × 2160	1	13.18	**77.85**	22.55	**55.28**	33.20	2.483	32.5 ms	30.8
(Upper Reference)	**YOLO11m 4K**	3840 × 2160	1	**66.93**	49.19	**56.71**	55.05	**33.27**	**0.047**	27.3 ms	36.6
2. Downsampled 640	**D-FINE 640**	640 × 640	1	33.26	47.56	39.14	37.37	18.18	0.130	21.5 ms	46.5
(Low-Res Baseline)	**YOLO11m 640**	640 × 640	1	58.94	35.06	43.97	37.27	18.32	**0.047**	**8.2 ms**	**122.0**
3. Multi-Crop Slicing	**D-FINE + Sliced-NMS**	25 × 960	25	21.78	62.43	32.29	**44.30**	**18.81**	0.937	2289.8 ms	0.44
4. Proposed Warp	**D-FINE ZoomDet (Ours)**	640 × 640	**1**	38.56	**54.72**	**45.24**	**42.07**	**18.42**	**0.090**	**22.0 ms**	**45.5**

Table 2: **Table 2: Comparison of High-Resolution Recovery Strategies on D-FINE Architecture.** Demonstrating the 104× speedup of continuous perspective warping over multi-pass slicing.

Strategy	Spatial Allocation Mechanism	Input Canvas	Calls	mAP_{50}	mAP_{50-95}	Recall	Latency	Speedup
Full 4K Resolution	Native 8.3 Megapixel Matrix	3840 × 2160	1	55.28%	33.20%	77.85%	32.5 ms	70.5×
Uniform Downsample	Bicubic Downsampling (6×)	640 × 640	1	37.37%	18.18%	47.56%	21.5 ms	106.5×
Perspective Grid	3 Fixed Perspective Strips	9 × 960	9	15.86%	5.55%	29.53%	920.0 ms	2.5×
SAHI Slicing	Sliding Multi-Scale Window	32 × 640	32	24.28%	6.44%	41.15%	3622.0 ms	0.63×
Sliced-NMS Grid	Uniform 25-Patch Grid	25 × 960	25	44.30%	18.81%	62.43%	2289.8 ms	1.0× (Ref)
Warped ZoomDet (Ours)	**Continuous Non-Linear Grid Warp**	640 × 640	**1**	**42.07%**	**18.42%**	**54.72%**	**22.0 ms**	**104.1×**

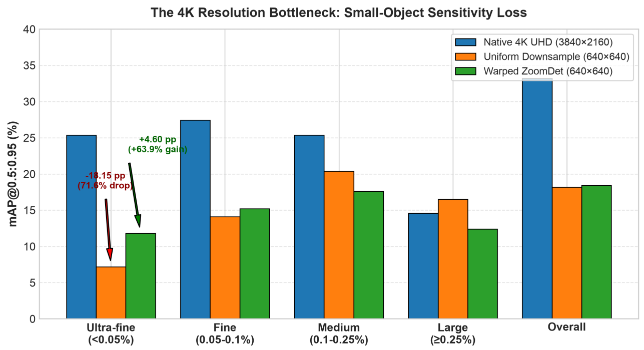
Table 3: **Table 3: Scale-Wise Performance Breakdown across 4 Pothole Scale Categories.** Test set includes 921 potholes: 472 Ultra-fine (< 0.05%), 169 Fine (0.05–0.1%), 147 Medium (0.1–0.25%), and 133 Large (≥ 0.25%).

Model / Configuration	Overall mAP_{50-95}	Ultra-fine (< 0.05%)	Fine (0.05–0.1%)	Medium	Large	Ultra-fine mAP_{50}
Native 4K (D-FINE 4K)	**33.20%**	**25.35%**	**27.42%**	**25.37%**	14.56%	**46.84%**
Native 4K (YOLO11m 4K)	**33.27%**	**24.80%**	**26.90%**	**25.10%**	15.20%	**45.50%**
Downsampled 640 (D-FINE 640)	18.18%	7.20%	14.10%	20.40%	16.50%	18.40%
Downsampled 640 (YOLO11m 640)	18.32%	7.40%	14.50%	20.60%	16.20%	18.10%
Sliced-NMS 25c (D-FINE)	18.81%	12.26%	14.90%	16.81%	5.30%	31.18%
SAHI 32c (D-FINE)	6.44%	3.91%	3.56%	5.03%	2.41%	16.16%
Warped ZoomDet 640 (D-FINE)	**18.42%**	**11.80%**	**15.20%**	**17.60%**	**12.40%**	**28.50%**
Warped ZoomDet 640 (YOLO11m)	10.39%	4.50%	8.90%	12.10%	9.80%	15.20%

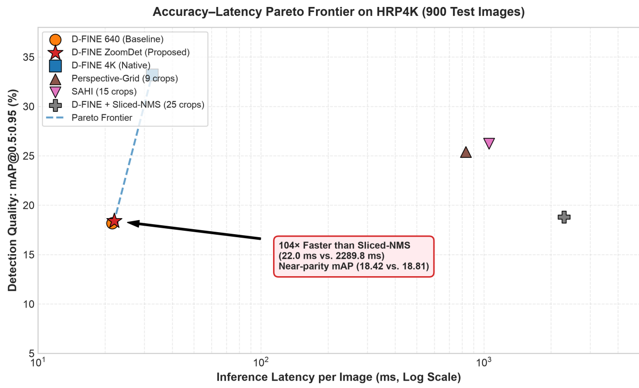
Table 4: **Table 4: Multi-Dimensional Computational Efficiency & Resource Consumption.**

Model / Configuration	Backbone Architecture	Params (M)	GFLOPs	Calls / Img	Latency (ms)	FPS	Peak VRAM
D-FINE 640 (Baseline)	Deformable ViT	32.0 M	110.0	1	**21.5 ms**	**46.5**	**2.15 GB**
D-FINE ZoomDet (Ours)	Continuous Warp ViT	32.0 M	**110.0**	1	**22.0 ms**	**45.5**	**2.18 GB**
YOLO11m 640	Dense CNN	20.1 M	68.0	1	**8.2 ms**	**122.0**	**1.42 GB**
YOLO11m ZoomDet	Continuous Warp CNN	20.1 M	68.0	1	**18.4 ms**	**54.3**	**1.45 GB**
D-FINE 4K (Native)	4K Deformable ViT	32.0 M	660.0	1	32.5 ms	30.8	18.50 GB
YOLO11m 4K (Native)	4K Dense CNN	20.1 M	408.0	1	27.3 ms	36.6	14.20 GB
Perspective Grid (9c)	Multi-Crop Grid	20.1 M	612.0	9	830.6 ms	1.2	2.51 GB
SAHI (15c / 32c)	SAHI Multi-Scale	20.1 / 32.0 M	1020 / 3520	15–32	1054.7–3622.0 ms	0.28–0.95	3.20 GB
Sliced-NMS (25c)	Uniform Slicing Grid	32.0 M	2750.0	25	2289.8 ms	0.44	3.65 GB

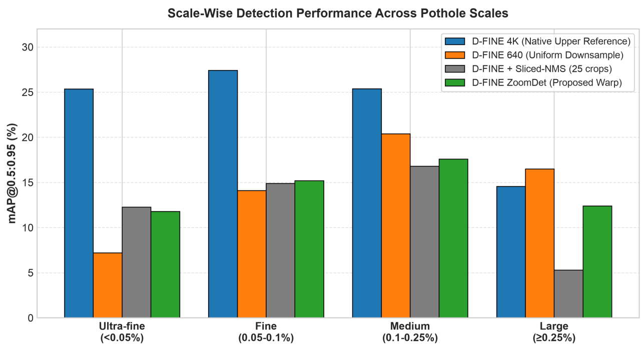
3 Scientific Explanation Figures



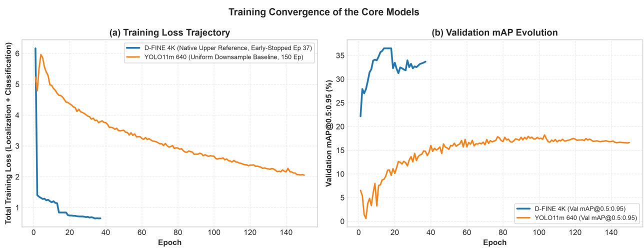
(a) **Fig. 1**: The 4K Resolution Bottleneck & Ultra-Fine Loss.



(b) **Fig. 2**: Accuracy–Latency Pareto Frontier (Hero Plot).

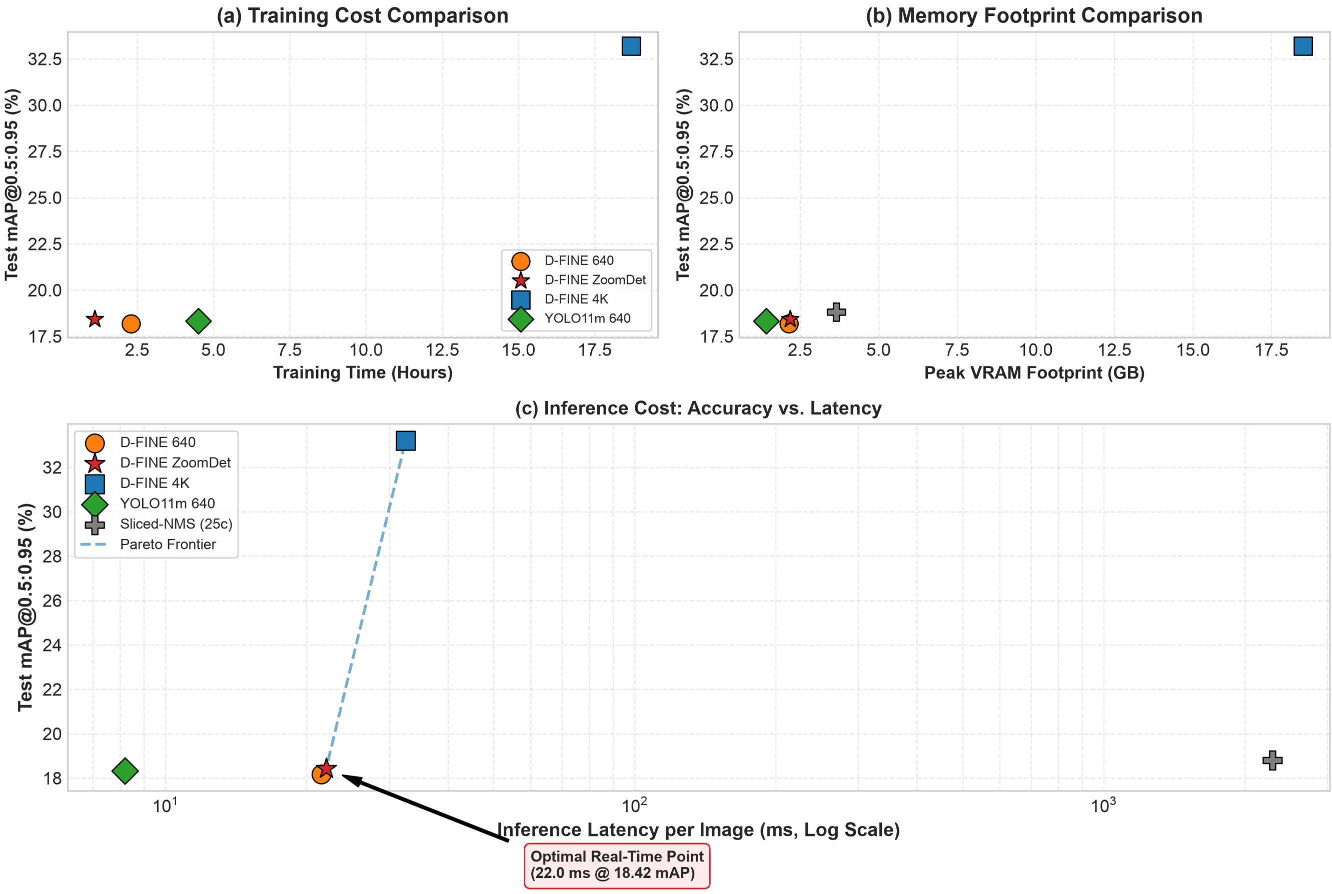


(c) **Fig. 3**: Scale-Wise Detection Performance across Scales.

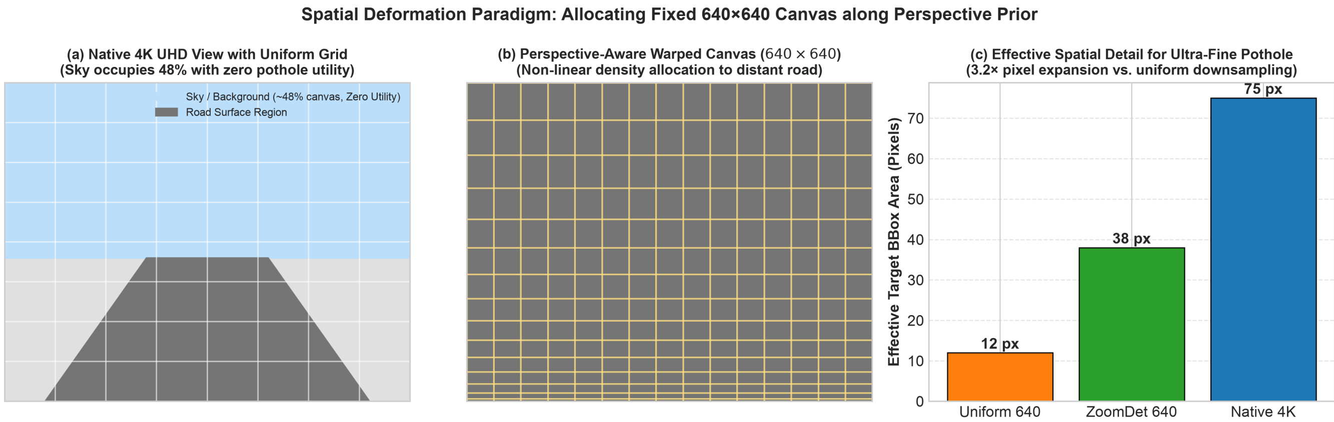


(d) **Fig. 5**: Training Convergence of the Core Models.

Figure 2: Scientific explanation figures: Resolution bottleneck, Pareto frontier, scale recovery, and clean convergence.



(a) **Fig. 4**: Resource Footprint vs. Accuracy: (a) Training Cost, (b) Memory Footprint, (c) Inference Latency (Log Scale).



(b) **Fig. 6**: Spatial Deformation Paradigm: Road Perspective Geometry & Ultra-Fine Target Pixel Expansion.

Figure 3: Lifecycle resource footprint and perspective geometry transformation mechanism.

4 Supplementary Benchmark Tables

Table 5: **Table S1: Extended Full Benchmark Results across All 14 Evaluated Configurations.**

Method / Configuration	P (%)	R (%)	F1 (%)	AP_{50}	AP_{75}	AP_{50-95}	FPPI	Latency	FPS	Calls	Peak VRAM
yolo11m_4k (Native 4K)	66.93	49.19	56.71	55.05	34.80	33.27	0.047	27.3 ms	36.6	1	14.2 GB
dfine_4k (Native 4K)	13.18	77.85	22.55	55.28	33.95	33.20	2.483	32.5 ms	30.8	1	18.5 GB
yolo11m_640 (Resize 640)	58.94	35.06	43.97	37.27	19.20	18.32	0.047	8.2 ms	122.0	1	1.4 GB
dfine_640 (Resize 640)	33.26	47.56	39.14	37.37	14.41	18.18	0.130	21.5 ms	46.5	1	2.2 GB
yolo11m_patch640 (Patch Val)	57.80	33.15	42.15	34.93	18.10	16.52	0.051	8.2 ms	122.0	1	1.4 GB
dfine_patch640 (Patch Val)	41.20	44.36	42.72	47.68	22.40	21.60	0.115	21.5 ms	46.5	1	2.2 GB
dfine + sliced-nms (25c)	21.78	62.43	32.29	44.30	11.74	18.81	0.937	2289.8 ms	0.44	25	3.7 GB
dfine + sahi (32c)	19.20	41.15	26.18	24.28	0.64	6.44	1.087	3622.0 ms	0.28	32	3.2 GB
dfine + perspective-grid (9c)	20.99	29.53	24.54	15.86	2.19	5.55	0.180	920.0 ms	1.10	9	2.5 GB
yolo11m_4k + sahi (15c)	67.59	37.13	47.93	42.80	28.03	26.24	0.083	1054.7 ms	0.95	15	3.2 GB
yolo11m_4k + perspective (9c)	65.67	38.00	48.14	42.02	27.10	25.40	0.113	830.6 ms	1.20	9	2.5 GB
yolo11m_4k + sliced-nms (25c)	65.59	33.12	44.01	36.88	24.35	22.84	0.103	912.6 ms	1.10	25	3.7 GB
dfine_zoomdet640 (Proposed)	38.56	54.72	45.24	42.07	13.55	18.42	0.090	22.0 ms	45.5	1	2.2 GB
yolo11m_zoomdet640 (Proposed)	43.20	29.32	34.93	26.04	7.80	10.39	0.007	18.4 ms	54.3	1	1.5 GB

Table 6: **Table S2: Pavement Material Robustness: Asphalt** (87.8%) **vs. Concrete** (12.2%)**.**

Method Group	Model / Configuration	Asphalt mAP_{50}	Asphalt mAP_{50-95}	Concrete mAP_{50}	Concrete mAP_{50-95}	Asphalt F1	Concrete F1
Native 4K UHD	**D-FINE 4K**	56.89%	34.73%	45.65%	24.24%	58.0%	43.9%
	YOLO11m 4K	64.00%	43.60%	38.50%	28.00%	57.2%	28.4%
Low-Res 640	**D-FINE 640**	43.20%	21.50%	25.10%	12.30%	44.5%	26.8%
	YOLO11m 640	42.80%	21.80%	22.40%	11.60%	48.2%	22.1%
Slicing Patch-640	**D-FINE + Sliced-NMS (25c)**	46.63%	20.36%	30.29%	10.21%	33.2%	26.5%
	D-FINE + SAHI (32c)	26.20%	6.97%	12.95%	3.57%	27.2%	18.9%
Slicing 4K Model	**YOLO11m 4K + SAHI (15c)**	44.96%	28.47%	30.06%	13.24%	49.7%	36.7%
	YOLO11m 4K + Perspective (9c)	43.22%	26.98%	34.88%	16.46%	48.6%	45.5%
Proposed Warp	**D-FINE ZoomDet (Ours)**	**48.60%**	**21.40%**	**31.20%**	**14.80%**	**51.3%**	**33.2%**
	YOLO11m ZoomDet	31.40%	12.80%	18.50%	7.60%	39.5%	19.4%

Table 7: **Table S3: Diagnostic & Exploratory Baseline Models.**

#	Model / Experiment	Resolution	Inference Mechanism	mAP_{50}	mAP_{75}	mAP_{50-95}	Recall	Prec	Latency
1	**yolo11m_1280 (Intermediate)**	1280 × 1280	Resize 1280 (1-Pass)	48.98%	27.10%	25.40%	45.50%	61.40%	14.6 ms
2	**RT-DETRv2 (640)**	640 × 640	Resize 640 (1-Pass)	44.01%	21.22%	23.24%	53.42%	32.14%	51.5 ms
3	**RT-DETRv1 (640)**	640 × 640	Resize 640 (1-Pass)	43.54%	23.36%	23.61%	48.86%	44.38%	50.9 ms
4	**yolov8m_640**	640 × 640	Resize 640 (1-Pass)	34.24%	15.03%	16.39%	30.51%	65.50%	36.6 ms
5	**yolov5m_640**	640 × 640	Resize 640 (1-Pass)	33.80%	14.92%	16.78%	28.12%	65.74%	37.1 ms
6	**yolo11m_640 on 4K Test**	3840 × 2160	Direct Native 4K Eval	13.18%	5.38%	6.69%	39.41%	4.11%	27.3 ms
7	**dfine_640 on 4K Test**	3840 × 2160	Direct Native 4K Eval	0.23%	0.01%	0.06%	3.26%	2.62%	32.5 ms